TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 10

5-MINUTE REVISION

Probability

Random experiments, events, the three approaches to probability, addition and multiplication laws, and Bayes' theorem.

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STAT 3202 · Chapter revision

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THE IDEA TO KEEP

Decide whether events overlap and whether they are independent.

REMEMBER

For a union, subtract the overlap. For an intersection, use a conditional probability unless independence is justified.

APPLY IT

If A and B are independent with probabilities 0.6 and 0.5, P(A and B) = 0.3 and P(A or B) = 0.8.

1 minute: key idea · 3 minutes: methods & examples · 1 minute: test yourself

1 Basic terms

Random experiment	Outcome cannot be predicted with certainty
Sample space S	Set of all possible outcomes
Sample point	One possible outcome
Event	A subset of the sample space
Random variable	Real-valued function of outcomes: discrete or continuous

2 Types of events

Mutually exclusive	Cannot occur together
Exhaustive	Union is the whole sample space
Equally likely	Equal probability
Independent	$P(A \cap B) = P(A)P(B)$
Complement A^c	All outcomes outside A
Impossible / certain	$P = 0$ / $P = 1$

3 Three approaches

Approach	Definition
Classical (mathematical)	$P(E) = \frac{m}{n}$: favourable ÷ total, for exhaustive, mutually exclusive, equally likely outcomes
Statistical (relative frequency)	$P(E) = \lim_{n\to\infty} \frac{m}{n}$ from repeated trials
Axiomatic (Kolmogorov)	(1) $P(E) \geq 0$ · (2) $P(S) = 1$ · (3) additivity for mutually exclusive events

Always $0 \leq P(E) \leq 1$ and $P(A^c) = 1 - P(A)$.

4 Set notation

$A \cup B$	A or B or both
$A \cap B$	Both A and B
A^c	Not A
$A \mid B$	A given B

5 Addition law

Any two events

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Mutually exclusive

$$P(A \cup B) = P(A) + P(B)$$

6 Conditional & multiplication

Conditional, $P(B) > 0$

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

General

$$P(A \cap B) = P(A)P(B \mid A)$$

Independent

$$P(A \cap B) = P(A)P(B)$$

With replacement: draws independent. Without replacement: generally dependent.

7 Bayes' theorem

Gives the probability of a **cause** given the observed **evidence**.

Two events

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Partition $E_1, \ldots, E_n$

$$P(E_i \mid A) = \frac{P(E_i)P(A \mid }{\sum_k P(E_k)P(A}$$

Prior

Probability before the evidence: $P(E_i)$

Posterior

Updated probability after the evidence: $P(E_i \mid A)$

8 Counting

Combinations (order ignored)

$$^{n}C_r = \frac{n!}{r!\,(n-r)!}$$

Gambler's fallacy: in independent trials, past outcomes do not change the next probability.

! Common mistakes

- ✗ Mutually exclusive ≠ independent: exclusive events with non-zero probabilities are always **dependent**.
- ✗ Forgetting to subtract $P(A \cap B)$ for events that can occur together · Confusing $P(A \mid B)$ with $P(B \mid A)$.

? Fill in the blanks

1. If P(A) = 0.3, P(B) = 0.5 and A and B are independent, P(A ∩ B) = ——. [0.15]
2. If P(A) = 0.4, P(B) = 0.5 and P(A ∩ B) = 0.2, then P(A ∪ B) = ——. [0.7]
3. If P(A ∩ B) = 0.12 and P(B) = 0.3, then P(A | B) = ——. [0.4]
4. A seed lot has 90% germination; the probability that a seed fails to germinate is ——. [0.1]
5. The number of ways to choose 2 varieties out of 5 is ——. [10]
6. Two mutually exclusive events, each with positive probability, cannot be ——. [independent]